

AKSHAY DUBEY

Mumbai, India

+91-8779197850 | akshaydubey1818@gmail.com | [akshay-dubey-2060](https://akshay-dubey-2060.github.io) | [akshaydubey50](https://akshaydubey50.github.io)

Summary

AI Full Stack Engineer with 4+ years of experience building production-grade applications at enterprise scale. Currently at **Fynd** (Reliance-backed), building an AI-powered hiring platform with multi-model LLM orchestration, RAG pipelines, and vector search. Builds and ships **LangGraph-based multi-agent systems** end-to-end – planning, tool-use, human-in-the-loop escalation, and full execution tracing – validated with real test suites and quantified evaluations, not demos.

Skills

AI/Agentic Systems: LangChain, LangGraph (multi-agent orchestration, ReAct reasoning patterns), RAG Pipelines (hybrid dense + BM25 retrieval, RRF fusion, LLM reranking), Vector Search (pgvector, ChromaDB), Prompt Engineering, Langfuse (LLM Observability), MCP (Model Context Protocol)

Frontend: React.js, Next.js, TypeScript, JavaScript, Redux, Zustand, RTK Query, Tailwind CSS

Backend: Node.js, Express.js, Python (FastAPI), GraphQL, REST APIs, Kafka, Celery, JWT Authentication

Databases: PostgreSQL, pgvector, MongoDB, Redis, Prisma ORM

Cloud & DevOps: GCP (Cloud Run), Docker, CI/CD Pipelines, Vercel, Supabase, Firebase

Practices & Tools: Agile/Scrum, Git, Sentry, PostHog, PG Boss, Cloudflare

Experience

Fynd (Shopsense Retail Technologies)

AI Full Stack Engineer

Mumbai, India

March 2025 – Present

HireFirst / Orion – AI Hiring Intelligence Platform

- Implemented **multi-model LLM council system** using OpenRouter coordinating GPT-4, Claude, and Gemini through a 3-stage pipeline: independent analysis, cross-model ranking, and chairman synthesis for enterprise hiring decisions.
- Extended **multi-vector RAG pipeline** using OpenAI text-embedding-3-large with 7 semantic vectors (skills, role, domain, seniority, functional, education, full-profile) stored in **pgvector** for candidate-job matching with weighted similarity scoring.
- Contributed to **asynchronous processing pipeline** using PG Boss queue with multi-stage orchestration (Parse, Deduplicate, Score), exponential backoff retry, and fuzzy deduplication for bulk candidate processing, in an **Agile/Scrum** delivery environment.
- Integrated **Langfuse for LLM observability** — enabling trace debugging across multi-model council sessions and token usage monitoring.
- Implemented tool-enabled AI agents using **MCP (Model Context Protocol)** with circuit breakers on AI/storage services for resilient document processing.
- Monitored and maintained the **GCP deployment pipeline**, performing deployment check-ins and diagnosing/fixing pipeline issues to keep releases reliable.

Fynd Commerce – E-commerce Platform (2,300+ brands, 20K+ stores)

- Developed full-stack **e-commerce bundle modules** using React.js, TypeScript, and **Python FastAPI**, building **REST and GraphQL APIs**, implementing pricing distribution algorithms, and integrating Kafka event streaming for commerce workflows.
- Built and enhanced **Python FastAPI** backend services, collaborating with frontend and backend teams to design scalable APIs, optimize request handling, and ensure seamless frontend-backend integration.
- Implemented **polygon-based geospatial shipping zones** using point-in-polygon calculations for hyperlocal delivery optimization.
- Served as **Maintenance DRI**, leveraging Sentry and PostHog to diagnose, troubleshoot, and resolve production issues across the retail platform while improving application reliability.

Osian Infotech Pvt Ltd

Software Developer

Mumbai, India

March 2022 – February 2025

- Built and maintained **Python FastAPI backend services** and REST APIs for enterprise applications – implementing business logic, request handling, and data flow between services – alongside React.js/TypeScript frontends for the same modules.
- Used **Redux Toolkit** and RTK Query for state/API caching, and React memoization/code-splitting for performance, improving Core Web Vitals across modules.

AI Agent Projects

AgentForge – Multi-Agent System with Human-in-the-Loop

LangGraph, FastAPI, Postgres, Redis, ChromaDB, Celery, Docker | github.com/akshaydubey50/agentforge

- Built a production-style **multi-agent system** in **LangGraph** – Supervisor plans, Specialists execute tools, a Reviewer validates – with retry-on-rejection, escalation to a human via API/dashboard, and **full execution tracing** (every node logs a `TraceSpan` for input, output, status, timing).
- Designed **persistent, resumable task state** in Postgres (not the in-memory graph checkpointer), enabling a paused task to be approved and resumed by an entirely different worker process, verified by an automated pause/resume integration test.
- Implemented sandboxed **agent tools** with real safety constraints: regex + transaction-level read-only SQL enforcement, network-disabled Docker code execution with enforced timeouts, and path-traversal-proof sandboxed file I/O.
- Added **long-term memory** (ChromaDB + Postgres) that ranks past task recall by a blend of semantic similarity and importance, informing future planning.
- Validated with **45 automated tests (43 passing, 2 gracefully-skipped)** against real Postgres, Redis, Chroma, Docker, and OpenAI calls – including a regression test that catches the system fabricating unretrieved data.

Hybrid RAG Eval – Hybrid-Search RAG with Quantified Evaluation

Python, FastAPI, ChromaDB, BM25, OpenAI | github.com/akshaydubey50/hybrid-rag-eval

- Built a **hybrid retrieval RAG pipeline** combining dense embeddings (OpenAI + ChromaDB) and sparse BM25 via **Reciprocal Rank Fusion**, followed by an **LLM reranking** stage over fused candidates.
- Implemented independent **citation verification** (a separate LLM pass checks each cited claim against source text) and a **multiplicative confidence score**, so an honest "no answer" scores zero confidence instead of a false-positive blend.
- Ran a hand-written **50-case golden evaluation set** across 3 chunking strategies (150 total evaluations, zero errors), achieving up to **98% answer correctness, 98%+ citation precision**, and **100% correct refusal** on out-of-scope questions.
- Shipped as a Dockerized service (FastAPI + Streamlit dashboard) with a repeatable eval harness (`run_eval.py`) so retrieval-strategy tradeoffs are measured, not assumed.

Education

Mumbai University

Bachelor of Science in Computer Science (BSc.CS); CGPA: 7.27

Maharashtra, India

July 2019 – March 2022